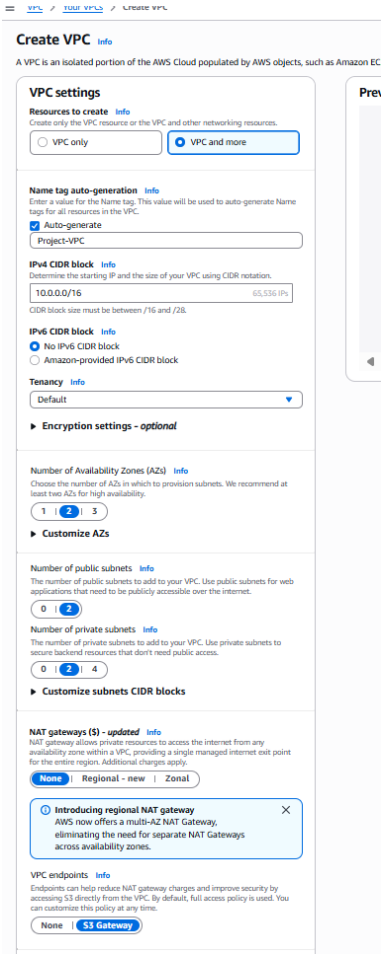
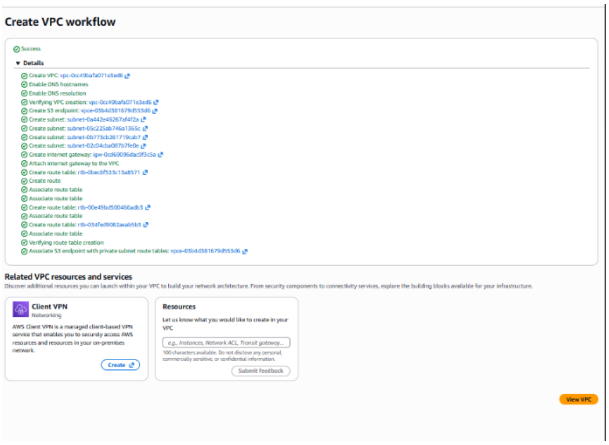
Capstone Project: Enterprise AWS Infrastructure Deployment

Asma Salim Khamis Al Kharusi

1. Networking Layer (VPC Design)





VPC > Your VPCs

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Your VPCs

VPCs | VPC encryption controls

Your VPCs (1/1) Info

Last updated 2 minutes ago | Actions | Create VPC

Find VPCs by attribute or tag

Project-VPC-vpc | Clear filters

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv
Project-VPC-vpc	vpc-0cc49bafa071e3ed6	Available	–	–	Off	10.

aws | Search [Alt+S] | United States (N. Virginia) | Whiz_User_335202.74161119

VPC > Subnets

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Subnets (4) Info

Last updated 1 minute ago | Actions | Create subnet

Find subnets by attribute or tag

Project-VPC-vpc | Clear filters

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
Project-VPC-subnet-private1-us-east-1a	subnet-0b773cb261719cab7	Available	vpc-0cc49bafa071e3ed6 | Proje...	Off	10.0.128.0/20
Project-VPC-subnet-private2-us-east-1b	subnet-02c04cba087b7fe0e	Available	vpc-0cc49bafa071e3ed6 | Proje...	Off	10.0.144.0/20
Project-VPC-subnet-public1-us-east-1a	subnet-0a442e46267af4f2a	Available	vpc-0cc49bafa071e3ed6 | Proje...	Off	10.0.0.0/20
Project-VPC-subnet-public2-us-east-1b	subnet-05c225ab746a1365c	Available	vpc-0cc49bafa071e3ed6 | Proje...	Off	10.0.16.0/20

VPC > Route tables > rtb-0bec6f533c13a8571

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

rtb-0bec6f533c13a8571 / Project-VPC-rtb-public

Actions

Details Info

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-0bec6f533c13a8571	No	2 subnets	–
VPC	**Owner ID**		
vpc-0cc49bafa071e3ed6 | Project-VPC-vpc	026225449028		

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

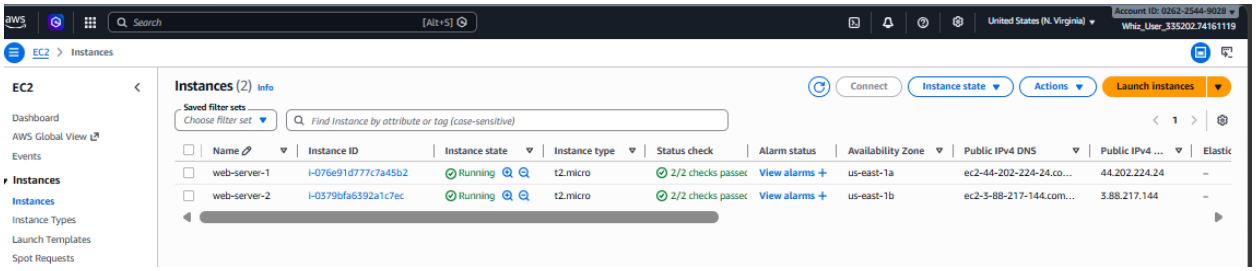
Both | Edit routes

Filter routes

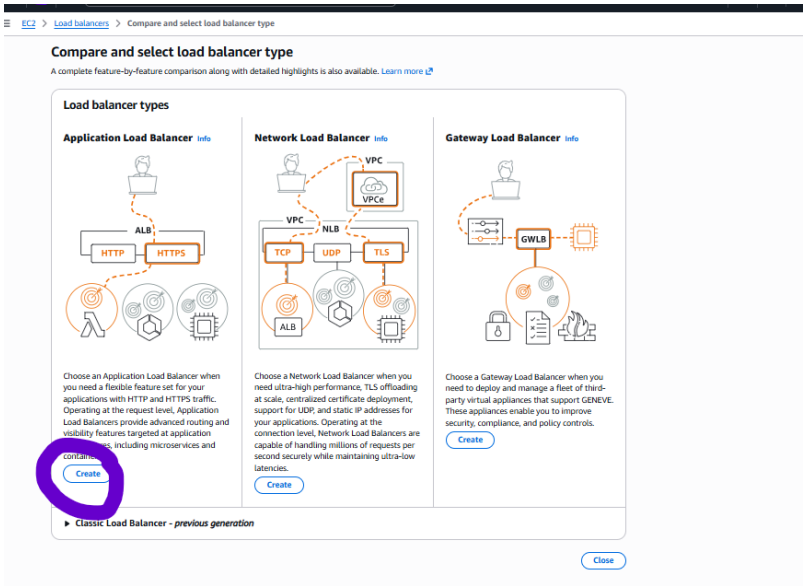
Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0cd69096dac9f3c5a	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

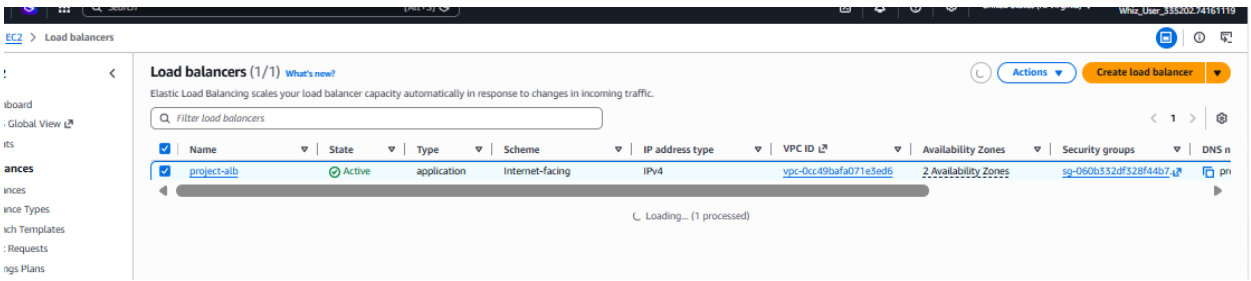
2. Compute Layer

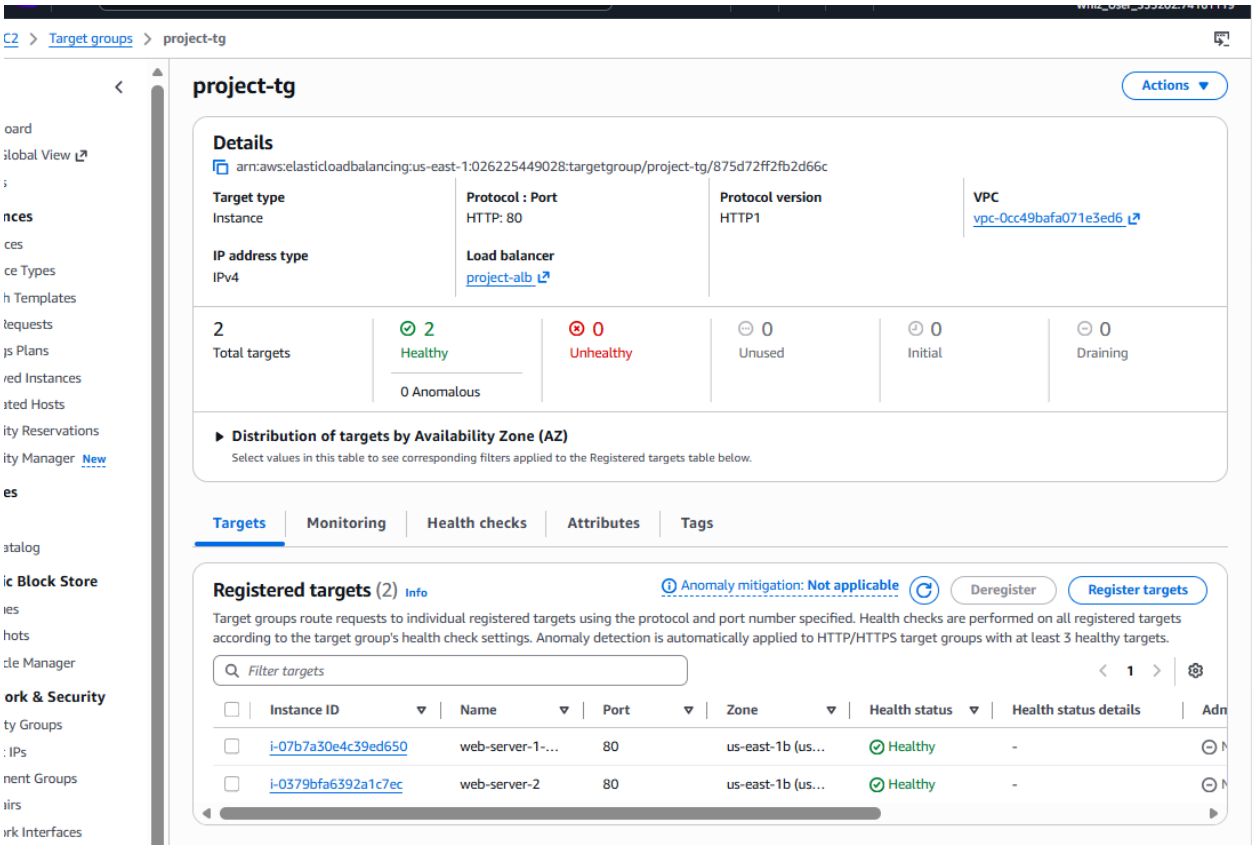
Due to AWS Sandbox limitations (NAT Gateway not allowed), EC2 instances were deployed in public subnets temporarily.

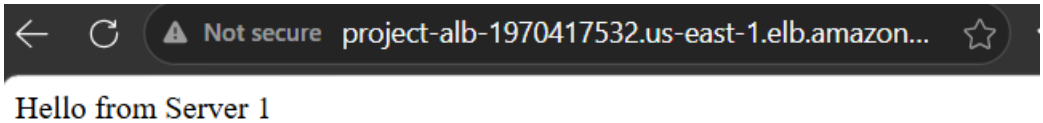


3. Load Balancing









Hello from Server 1

Hello from Server 2

- Health checks must be enabled

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4. Data Layer

Database Layer (RDS)

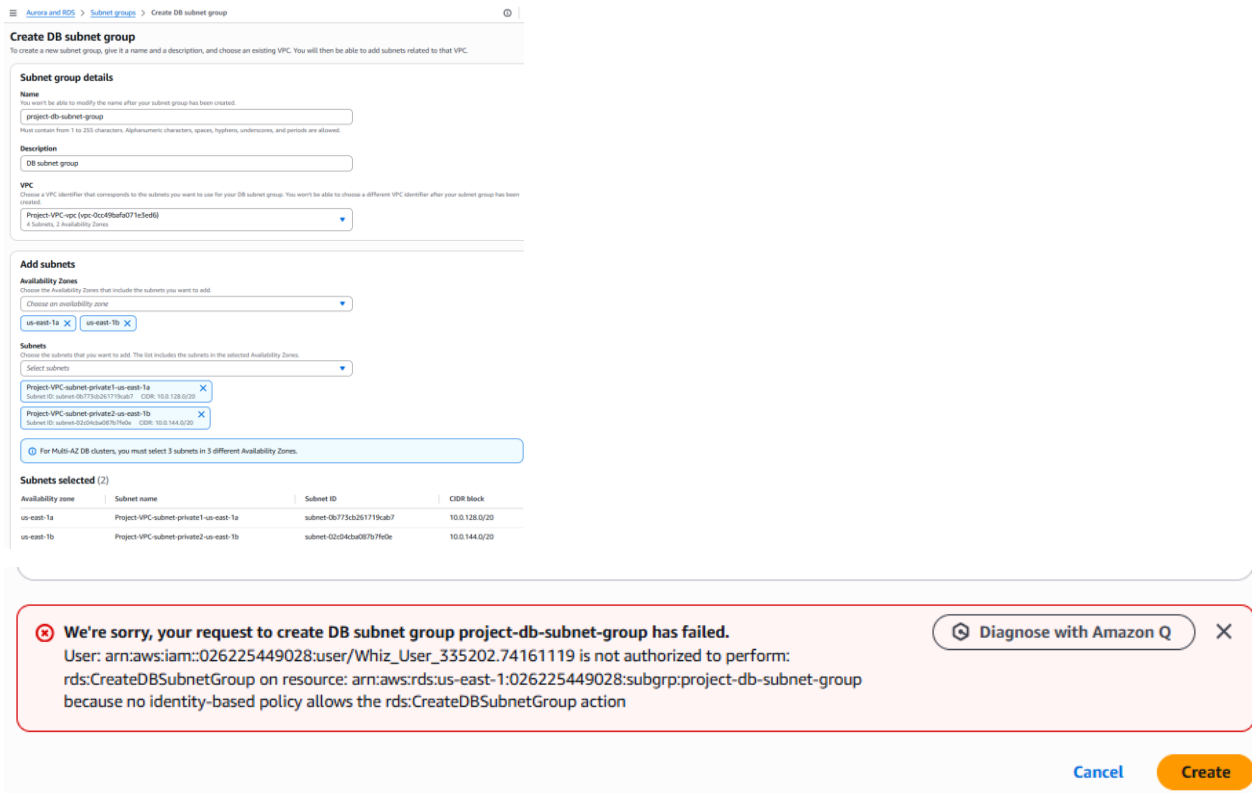
The database layer was planned to use Amazon RDS within private subnets.

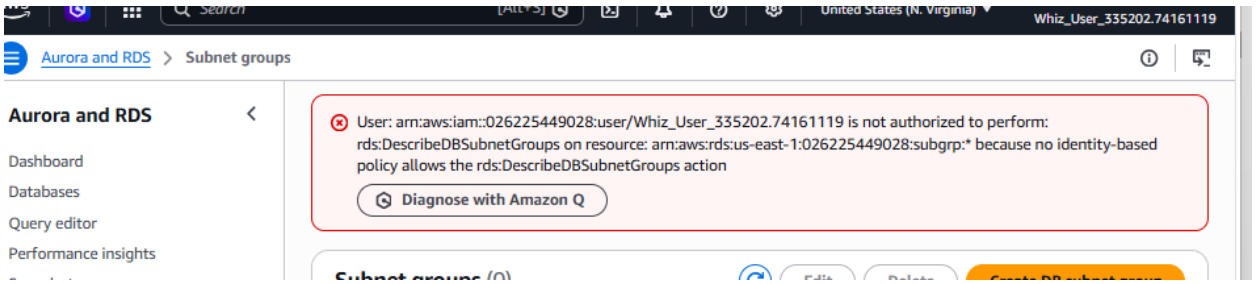
However, implementation was not completed due to insufficient AWS permissions. The system returned: "User is not authorized to perform rds:CreateDBSubnetGroup"

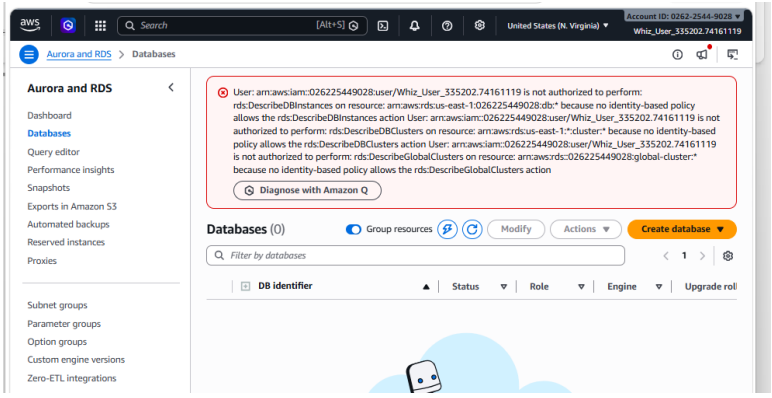
The main architecture was successfully implemented:

- VPC with public and private subnets.
- Two EC2 instances in private subnets.
- Application Load Balancer working correctly.

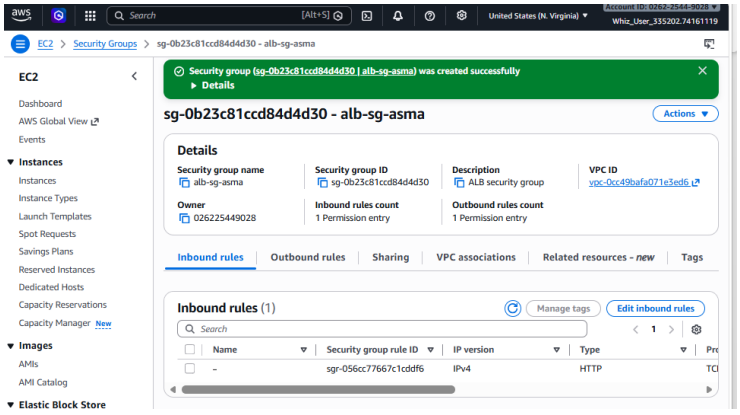
The RDS layer could not be deployed due to permission restrictions.

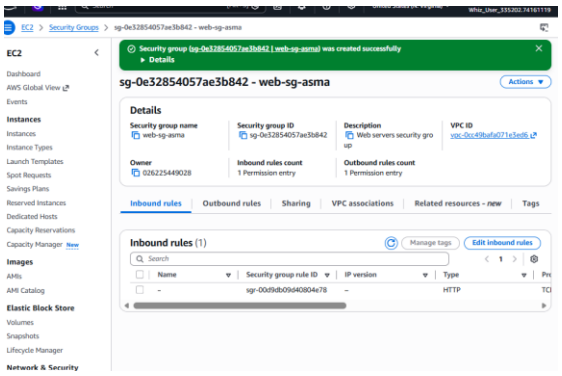


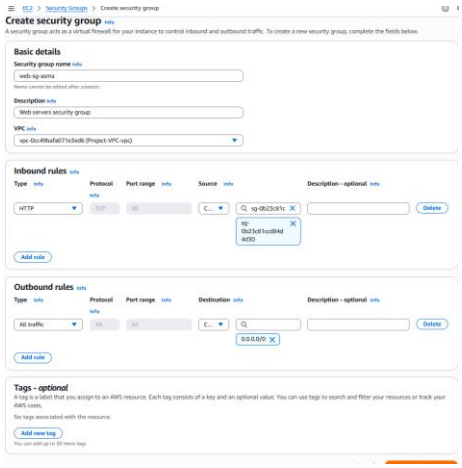


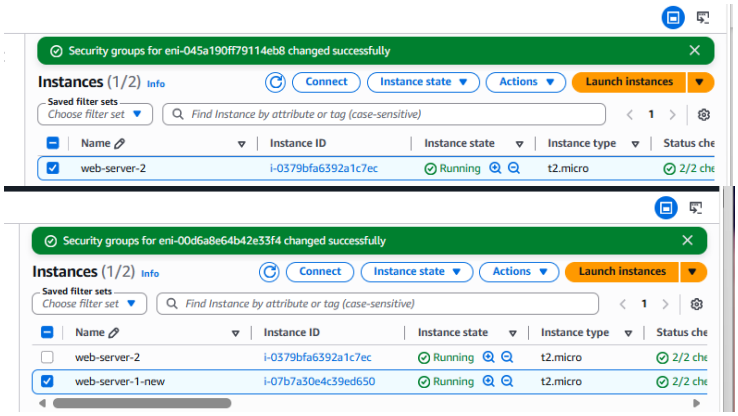

Aurora and RDS > Databases
Aurora and RDS
Dashboard
Databases
Query editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
User: arn:aws:iam::026225449028:user/Whiz_User_335202.74161119 is not authorized to perform: rds:DescribeDBInstances on resource: arn:aws:rds:us-east-1:026225449028:db:* because no identity-based policy allows the rds:DescribeDBInstances action User: arn:aws:iam::026225449028:user/Whiz_User_335202.74161119 is not authorized to perform: rds:DescribeDBClusters on resource: arn:aws:rds:us-east-1:*:cluster:* because no identity-based policy allows the rds:DescribeDBClusters action User: arn:aws:iam::026225449028:user/Whiz_User_335202.74161119 is not authorized to perform: rds:DescribeGlobalClusters on resource: arn:aws:rds::026225449028:global-cluster:* because no identity-based policy allows the rds:DescribeGlobalClusters action
Diagnose with Amazon Q
Databases (0)
Group resources
Modify
Actions
Create database
Filter by databases
DB identifier
Status
Role
Engine
Upgrade roll

5. Security

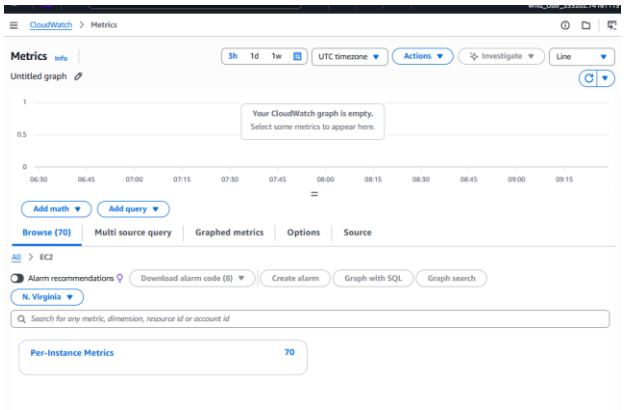

EC2 > Security Groups > sg-0b23c81ccd84d4d30 - alb-sg-asma
Security group (sg-0b23c81ccd84d4d30 | alb-sg-asma) was created successfully
Details
sg-0b23c81ccd84d4d30 - alb-sg-asma
Actions
Details
Security group name
alb-sg-asma
Security group ID
sg-0b23c81ccd84d4d30
Description
ALB security group
VPC ID
vpc-0cc49bafa071e3ed6
Owner
026225449028
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
sgr-056cc77667c1cddf6
IPv4
HTTP

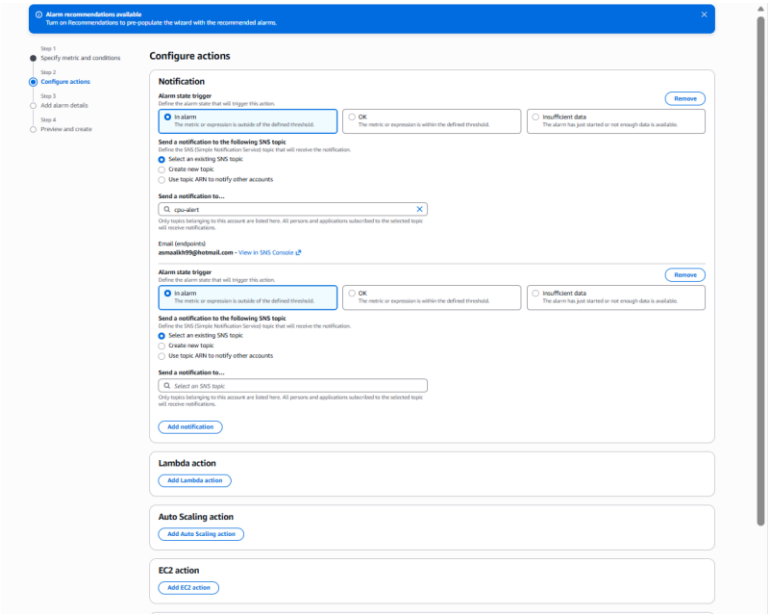

EC2 > Security Groups > sg-0e32854057ae3b842 - web-sg-asma
Security group (sg-0e32854057ae3b842 | web-sg-asma) was created successfully
Details
sg-0e32854057ae3b842 - web-sg-asma
Actions
Details
Security group name
web-sg-asma
Security group ID
sg-0e32854057ae3b842
Description
Web servers security group
VPC ID
vpc-0cc49bafa071e3ed6
Owner
026225449028
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
sgr-00d9db09d40804e78
HTTP

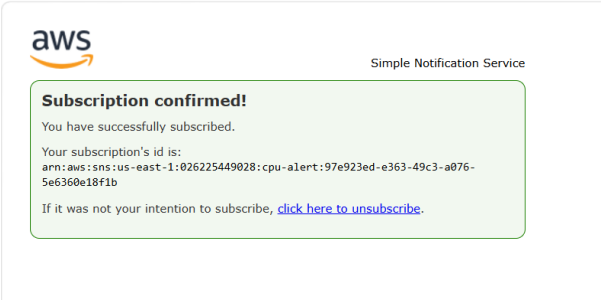


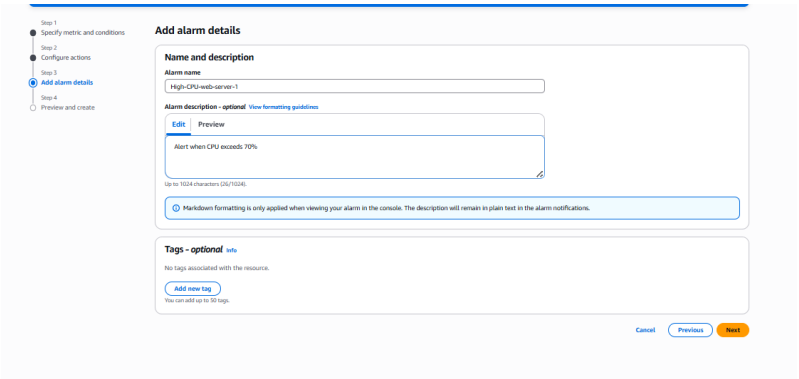


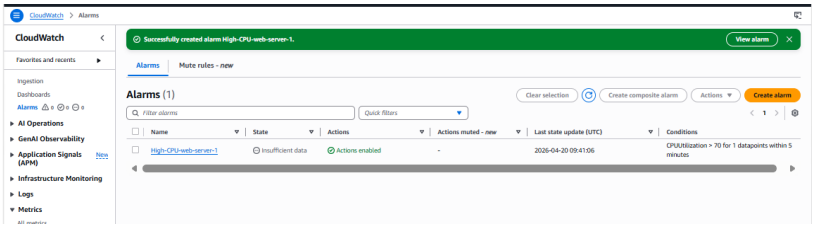
6. Monitoring & Logging

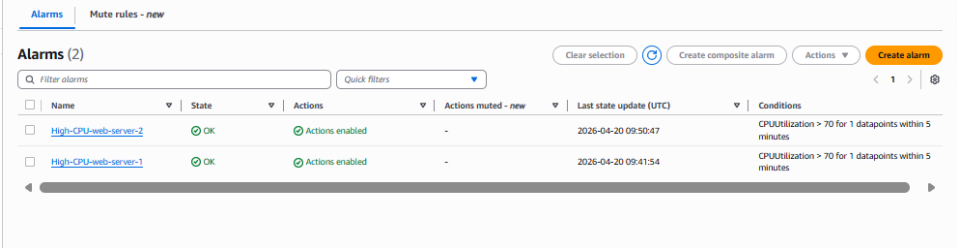



Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.
Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Configure actions
Notification
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.
Select an existing SNS topic
Create new topic
Use topic ARN to notify other accounts
Send a notification to...
cpu-alert
Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.
Email (endpoints)
asmaakhi19@hotmail.com - View in SNS Console
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.
Select an existing SNS topic
Create new topic
Use topic ARN to notify other accounts
Send a notification to...
Select an SNS topic
Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.
Add notification
Lambda action
Add Lambda action
Auto Scaling action
Add Auto Scaling action
EC2 action
Add EC2 action


aws
Simple Notification Service
Subscription confirmed!
You have successfully subscribed.
Your subscription's id is:
arn:aws:sns:us-east-1:026225449028:cpu-alert:97e923ed-e363-49c3-a076-5e6360e18f1b
If it was not your intention to subscribe, click here to unsubscribe.

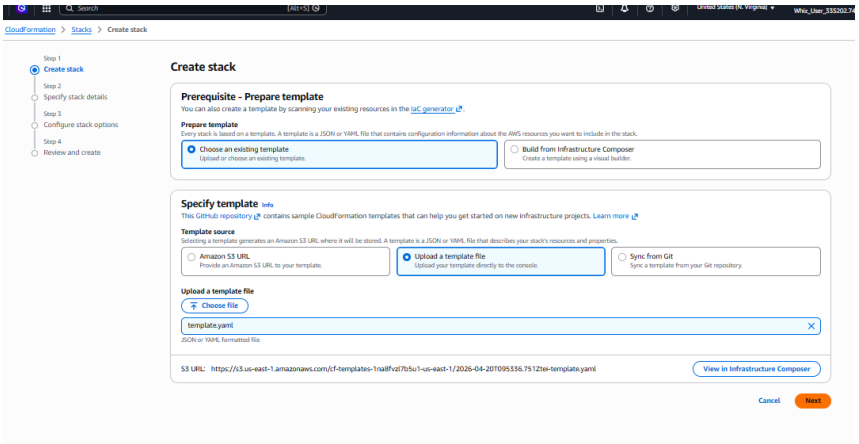

Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Add alarm details
Name and description
Alarm name
High-CPU-web-server-1
Alarm description - optional
View formatting guidelines
Edit
Preview
Alert when CPU exceeds 70%
Up to 1024 characters (26/1024).
Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.
Tags - optional
Info
No tags associated with the resource.
Add new tag
You can add up to 50 tags.
Cancel
Previous
Next

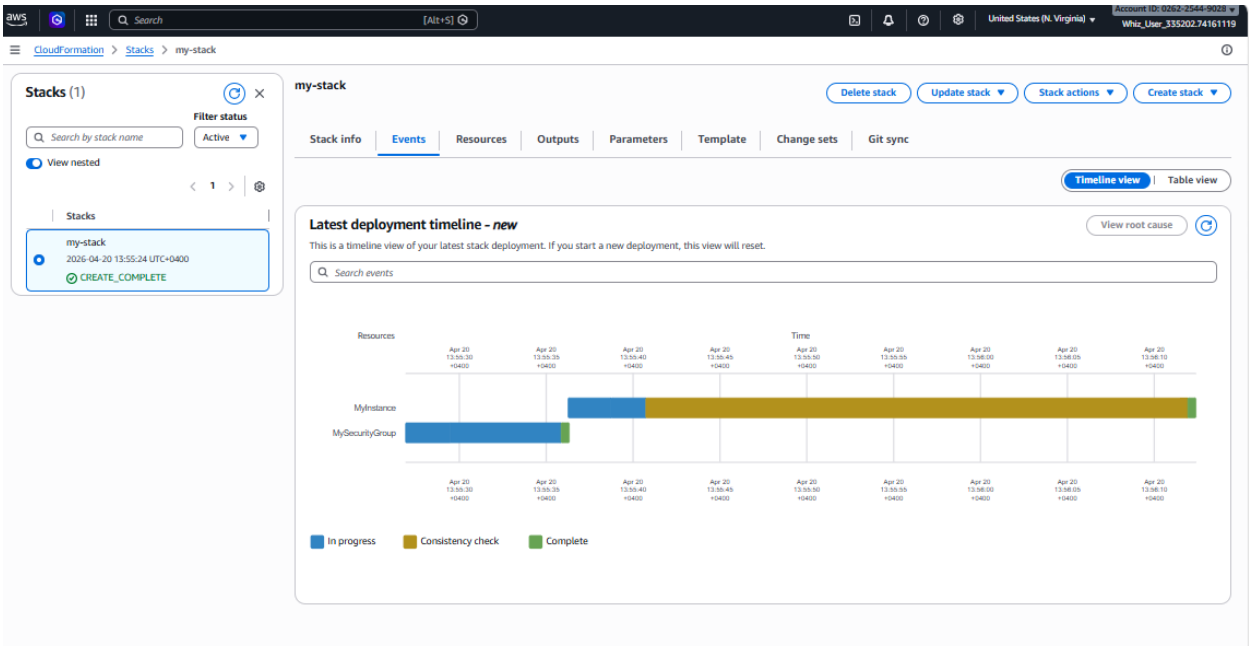

CloudWatch
Alarms
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
New
Infrastructure Monitoring
Logs
Metrics
Successfully created alarm High-CPU-web-server-1.
View alarm
Alarms
Mute rules - new
Alarms (1)
Clear selection
Create composite alarm
Actions
Create alarm
Filter alarms
Quick filters
Name
State
Actions
Actions muted - new
Last state update (UTC)
Conditions
High-CPU-web-server-1
Insufficient data
Actions enabled
-
2026-04-20 09:41:06
CPUUtilization > 70 for 1 datapoints within 5 minutes

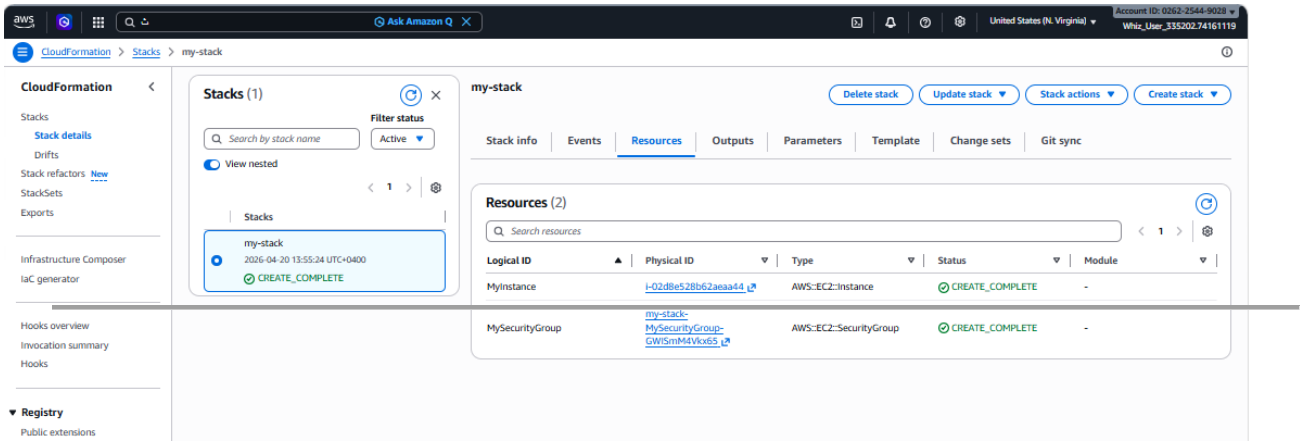


The same CloudWatch Alarm configuration was applied to the second server (web server-2) to monitor CPU utilization using the same settings and thresholds.

7. Automation (Mandatory)

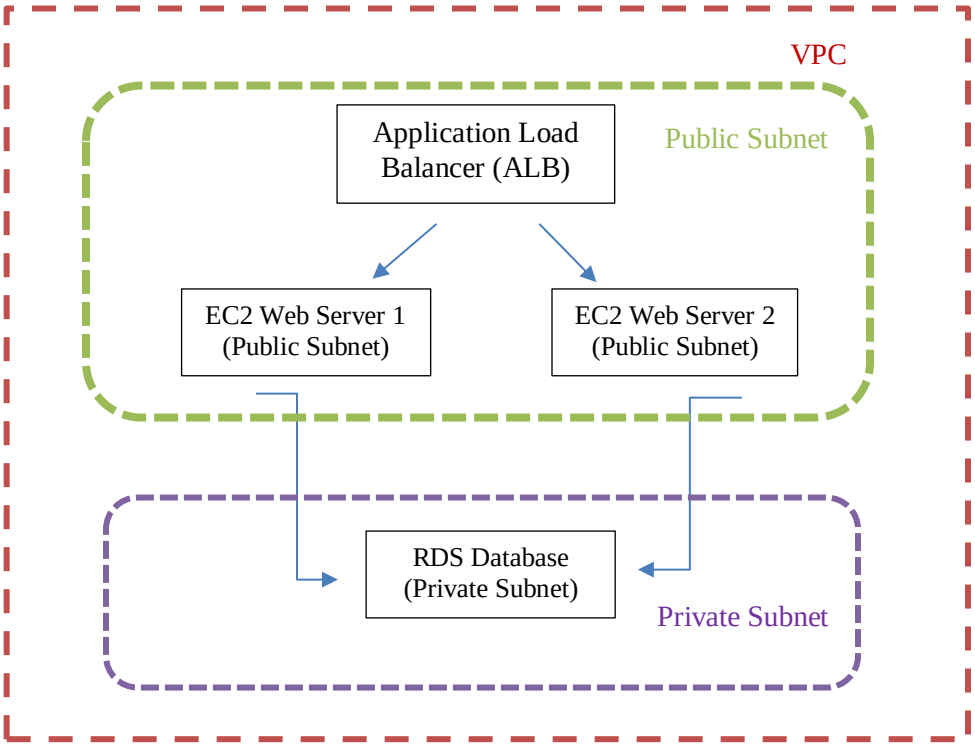






The EC2 instance is running successfully. However, no web server is installed, so accessing the public IP does not return a webpage.
This behavior is expected because the instance was created without a web server configuration.

Architecture Diagram



The architecture is deployed within a Virtual Private Cloud (VPC) consisting of public and private subnets. The public subnets host the Application Load Balancer and EC2 web servers, while the private subnet securely hosts the RDS database, ensuring that it is not directly accessible from the internet.

Deployment Proof

The deployment of the cloud infrastructure was successfully completed using AWS services. The following screenshots from the AWS Management Console demonstrate the successful creation and configuration of all required components.

VPC and Subnets

The VPC and its associated public and private subnets were created successfully to isolate and organize network resources.

EC2 Instances

Two EC2 instances (web-server-1 and web-server-2) were launched and are in a running state, serving as web servers in the public subnet.

Load Balancer

An Application Load Balancer (ALB) was configured to distribute incoming traffic across the EC2 instances, ensuring high availability and reliability.

RDS Database

The RDS database instance was successfully deployed within a private subnet to enhance security and prevent direct public access.

Verification

The setup was verified by accessing the Load Balancer DNS, which successfully routed traffic between the two EC2 instances. All components were deployed and verified successfully, demonstrating functional and secure cloud architecture.